



MSE 718

Prof. Jangho Yang

Toronto Municipal Building

Energy Efficiency Analysis, 2017-2024

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1. Introduction

As cities intensify climate action, buildings have become a central decarbonization target. In Toronto, a large share of GHG emissions comes from building heating and operations, and the TransformTO Net Zero Strategy seeks to reduce both energy demand and reliance on fossil fuels such as natural gas.

However, a building type that appears inefficient in raw data may simply be larger, operate longer, or serve more intensive functions. This matters because decarbonization depends not only on using less energy, but also on distinguishing abnormal inefficiency from expected operational scale.

Using Toronto's municipal energy disclosure data from 2017–2024, this study asks: after accounting for building size, operating schedule, and year effects, which building types systematically deviate from the portfolio baseline in energy use intensity (EUI) and GHG intensity? By separating raw differences from adjusted outcomes, we examine whether the energy and carbon stories diverge and provide evidence for more targeted decarbonization investment.

2. Methods

2.1 Data Cleaning and Data Construction

We constructed an unbalanced panel dataset of Toronto municipal buildings spanning 2017–2024 (excluding 2021 due to substantial missing data) using publicly available annual energy disclosure records (City of Toronto, n.d.-a). Although the database began in 2011, we restrict the sample to 2017 onward to align with the TransformTO policy era.

The analytical sample is limited to six municipal building types—Fire Stations, Libraries, Offices, Community Centres, Recreation Facilities, and Police Stations—to ensure comparability. Records with gross floor area below 1,000 square feet are excluded, as these likely represent non-building assets or would mechanically inflate intensity measures.

Buildings are matched across years using Portfolio Manager IDs for 2022–2024 and standardized address–postal code combinations for earlier years, with the cleaned 2024 dataset serving as the reference sample. To ensure cross-year comparability, we retain only electricity and natural gas, the two most consistently reported energy sources. This choice also aligns with the study's focus on operational decarbonization, since natural gas is a major source of building-level carbon emissions. Annual energy use is aggregated to the building-year level where necessary.

Missing energy values are set to zero only when they plausibly indicate non-use, based on consistent reporting patterns or substitution by another energy source. Floor area is standardized to square feet, and EUI and GHG intensity are constructed using consistent conversion and emission factors (Environment and Climate Change Canada, 2024, Table

A13–7). Extreme outliers are trimmed, and log transformations are applied. Descriptive summaries can be found in Section 6.3 and 6.4.

2.2 Model Specification

To examine systematic differences in building performance, we estimate Bayesian multilevel regression models for two outcomes: log-transformed energy use intensity (EUI) and log-transformed GHG intensity, each comprising three nested specifications:

Model 1 — Baseline (no multilevel structure):

$$\log(Y_i) = \alpha + \beta_1 \log(GFA_i) + \beta_2 \log(Hours_i) + \beta_3 Year_i + \varepsilon_i$$

Model 2 — Random Intercept:

$$\begin{aligned} \log(Y_i) &= \alpha_{[i]} + \beta_1 \log(GFA_i) + \beta_2 \log(Hours_i) + \beta_3 Year_i + \varepsilon_i \\ \alpha_{[i]} &\sim Normal(\mu\alpha, \sigma\alpha) \end{aligned}$$

Model 3 — Random Intercept + Slope:

$$\begin{aligned} \log(Y_i) &= \alpha_{[i]} + \beta_1_{[i]} \log(GFA_i) + \beta_2 \log(Hours_i) + \beta_3 Year_i + \varepsilon_i \\ (\alpha_{[i]}, \beta_1_{[i]})^T &\sim MVNormal((\mu\alpha, \mu\beta_1)^T, \Sigma) \end{aligned}$$

where Y_i denotes either EUI or GHG intensity for observation i , $j[i]$ indexes the building type of observation i , and $\log(GFA)$ denotes log gross floor area, and $\log(Hours)$ denotes log weekly operating hours. Detailed model comparison results can be found in Supplementary Information Table A1. Year was included as a scaled continuous variable ($Year_scaled$) to capture common temporal trends while improving numerical stability in estimation.

2.3 Priors

We assigned weakly informative priors throughout. For the energy track, the global intercept received $Normal(3, 1)$, reflecting that a building of average size and operating hours has $\log(EUI)$ approximately 3, corresponding to approximately 20 ekWh/sqft — consistent with the observed portfolio mean. For the carbon track, the intercept received $Normal(1, 1)$, corresponding to approximately 2.7 kg CO₂e/sqft. All slope coefficients received $Normal(0, 1)$, group-level standard deviations received $Exponential(1)$, and the correlation matrix for Model 3 received $LKJ(2)$, placing mild shrinkage toward independence. Prior predictive checks (see Figure A3) confirmed that these priors generate physically plausible EUI and GHG values prior to observing data. All priors are specified on the log-transformed scale of the outcome variables.

2.4 Estimation Approach

We adopt a partial pooling strategy through Bayesian multilevel model rather than estimating each building type independently. Estimating separate models by type would produce unstable results for smaller groups and fail to leverage shared information across categories.

The multilevel framework allows building types to have their own intercepts, capturing systematic differences in baseline performance. In the more flexible E3/C3 specification, building types are additionally allowed to have their own floor-area slopes, enabling the relationship between building size and energy or carbon intensity to vary across categories.

Partial pooling improves estimation stability by shrinking group-level estimates toward the overall mean in proportion to group size. Model comparison using LOO-CV shows that the varying-intercept, varying-slope specification (E3/C3) provides superior predictive performance.

2.5 Model Checks and Diagnostics

All models were estimated using Hamiltonian Monte Carlo via brms, with four chains and 2,000 iterations (1,000 warmup). Model adequacy was assessed using R-hat statistics, effective sample size measures, trace plots, prior and posterior predictive checks.

We treat E3/C3 as the primary models because they perform best under LOO-CV. Their R-hat values are extremely close to 1, and trace plots show broadly stable mixing, although a small number of divergent transitions remain after warmup. Posterior predictive checks indicate that the fitted models reasonably reproduce the overall centre and broad shape of the observed distributions, with mild mismatch in the tails.

The simpler random-intercept models (E2/C2) are retained as benchmark specifications, as shown in the appendix. They provide a more constrained comparison with common floor-area slopes across building types.

3. Results

3.1 Model Comparison

Model comparison is conducted using leave-one-out cross-validation (LOO-CV). Across both the energy and carbon specifications, the varying-intercept, varying-slope models (E3/C3) consistently outperform the simpler random-intercept models (E2/C2) in terms of expected log predictive density.

This indicates that allowing the effect of floor area to vary across building types improves model fit and better captures structural heterogeneity in the data. Accordingly, E3/C3 are adopted as the primary models, while E2/C2 are retained as benchmarks.

3.2 Building-Type Differences

Figure 2 reports the adjusted E3/C3 results. After controlling for floor area, operating hours, and year, Fire Stations remain the clearest above-baseline category, especially on the carbon track, indicating that their elevated intensity cannot be explained simply by larger size or longer operating schedules. Recreation, by contrast, falls below the carbon baseline after

adjustment, suggesting that its high raw intensity reflects scale and usage rather than persistent inefficiency. Office buildings no longer remain clearly below the carbon baseline once slope heterogeneity is introduced, indicating that the earlier benchmark result is not fully robust.

Overall, the adjusted results show that carbon differences are more policy-informative than raw EUI rankings, reinforcing the importance of fuel mix rather than total energy use alone.

3.3 Residual Analysis

To complement the building-type-level findings, we conduct a residual analysis to identify buildings that systematically underperform relative to model expectations. Residuals are computed using the E2/C2 models, which serve as benchmark specifications due to their simpler random-intercept structure and greater stability for building-level comparison.

Residuals are defined as the difference between observed and predicted $\log(\text{EUI})$ and $\log(\text{GHG intensity})$, after controlling for building type, floor area, operating hours, and year effects. Positive residuals indicate above-expected energy use or emissions. Under the modeling framework, these residuals can be interpreted as deviations from expected performance conditional on observed covariates, and therefore serve as a proxy for relative inefficiency.

In the 2024 cross-section, Pelmo Park and Wellesley Community Centre emerge as the top underperformers on both energy and carbon tracks. Both also appear consistently across multiple years, suggesting persistent inefficiencies rather than temporary variation. The Forensic Service, Store & Garage (Police Station) shows a similar pattern, aligning with the type-level finding of elevated carbon intensity for Police Stations.

Overall, the results highlight substantial within-type heterogeneity, indicating that policy should combine category-level prioritization with building-level interventions.

4. Discussion

This study provides a model-based empirical assessment of building performance, revealing a clear divergence between energy use and carbon outcomes. While no credible differences are observed in EUI after adjustment, substantial differences emerge in GHG intensity, indicating that carbon outcomes are driven primarily by energy composition rather than total energy use.

4.1 Interpreting Raw Differences in Energy and Carbon Performance

At first glance, the raw distributions suggest that some building types are clearly more energy-intensive than others. As shown in **Figure 1**, Recreation and Fire Station appears higher in the unadjusted EUI distribution, while Office appears lower. However, these visible raw differences combine multiple influences—including building size, operating schedule, and functional demand—and therefore should not be interpreted as direct evidence of

abnormal inefficiency. Once these factors are incorporated into the Bayesian multilevel models, the picture changes. **Figure 2** shows that Fire Station remains the clearest above-baseline category, especially on the carbon track, whereas Recreation becomes a reversal case: although it looks high in the raw data, it does not remain unusually inefficient after adjustment. This distinction is important because it shows that raw rankings alone can be misleading, and that the more policy-relevant question is which building types remain above baseline after controlling for key structural and operational factors.

Figure 1. EUI Distribution by Building Type

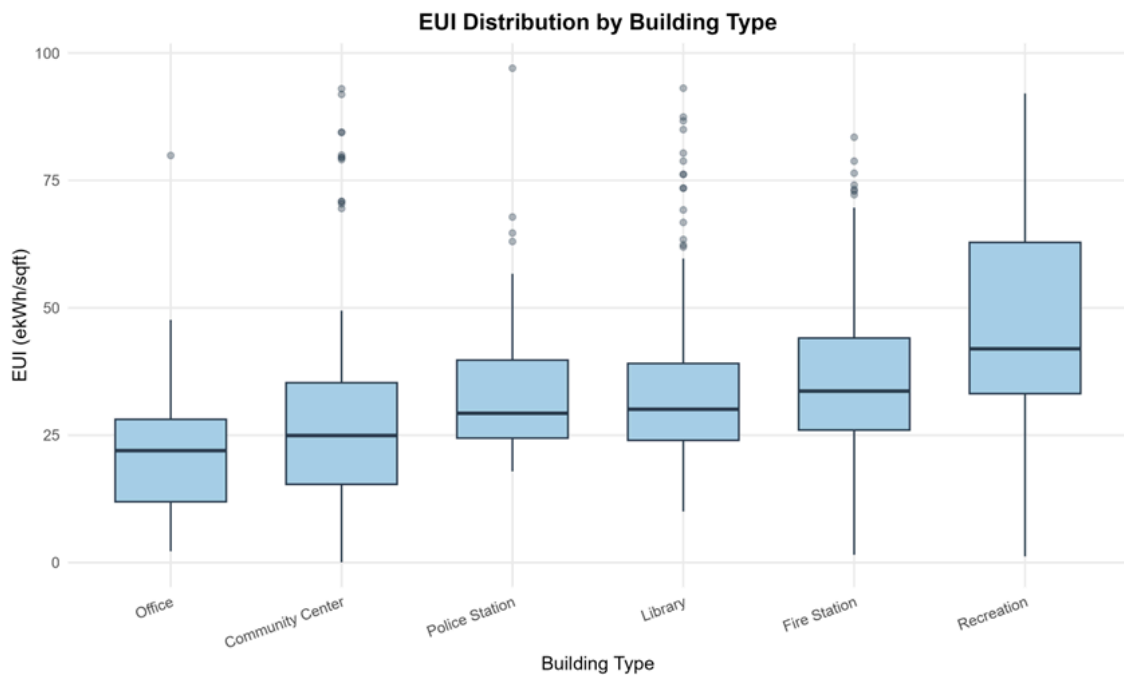
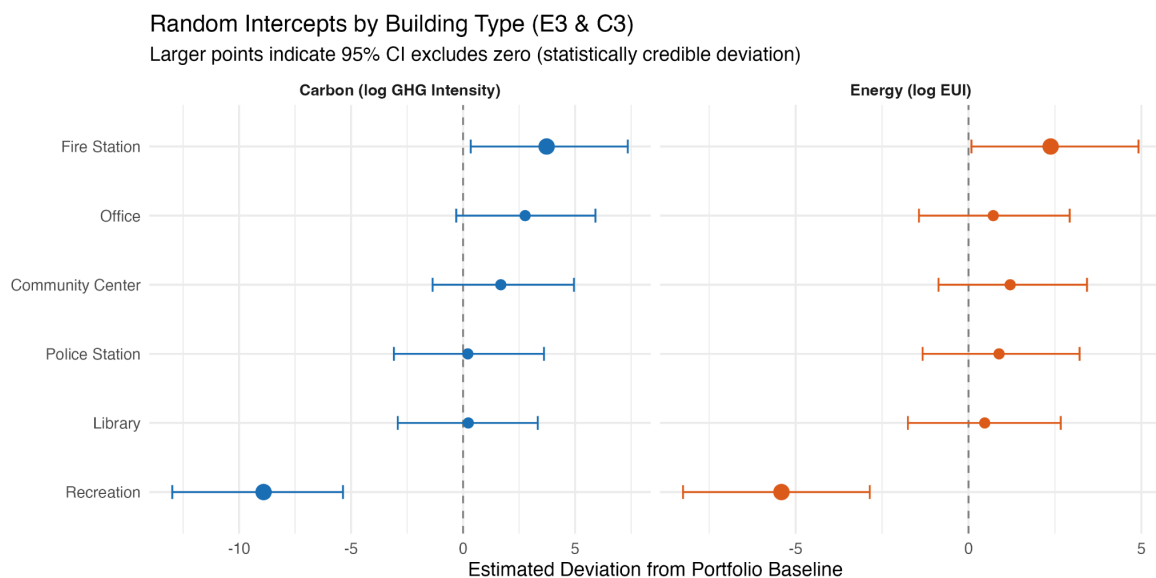


Figure 2. Random Intercepts by Building Type (E3/C3)



4.2 Policy Implications for Toronto's Net Zero Strategy

These results matter for Toronto's decarbonization strategy because raw EUI rankings alone would suggest targeting the most energy-intensive categories in descriptive data. The adjusted results point to a narrower policy problem: the more meaningful differences appear on the **carbon** track, where **Fire Stations** remain more clearly above baseline than in the energy results alone.

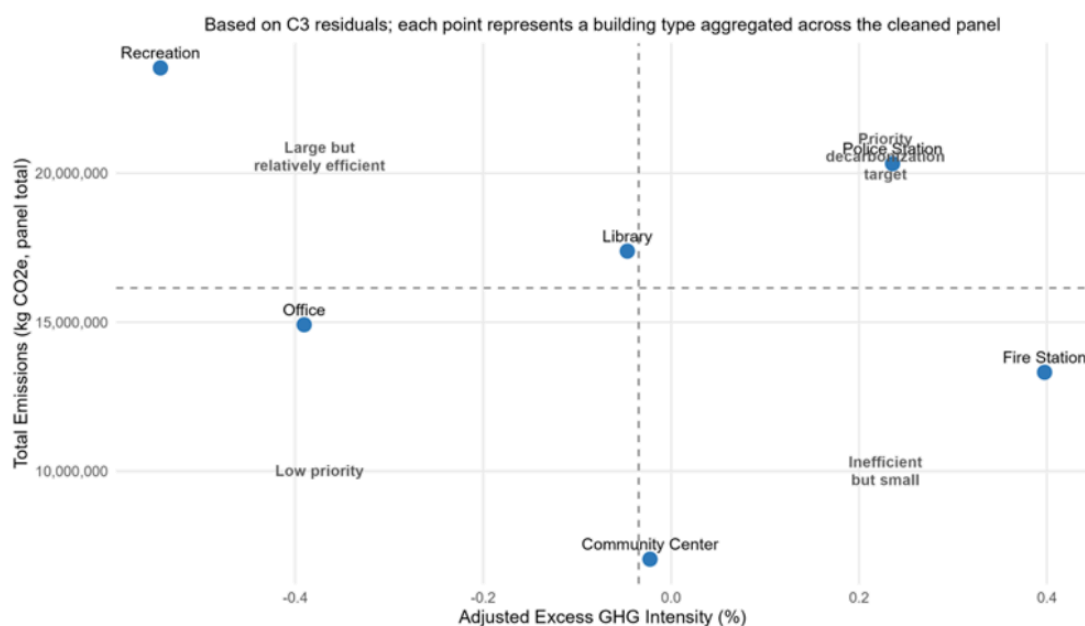
This suggests that the main challenge is not only reducing energy use, but also changing the **composition of energy use**. Categories that remain elevated in adjusted carbon intensity are more likely to reflect reliance on carbon-intensive systems, especially fossil-fuel-based heating. The policy implication is therefore a shift from general efficiency improvement toward targeted decarbonization through electrification, fuel switching, and building-system transition.

4.3 Prioritization Under Budget Constraints

Figure 3 compares adjusted excess carbon intensity with total emissions. This separates categories whose emissions are mainly driven by scale from those that remain carbon-intensive after controlling for floor area, operating hours, and year. Under budget constraints, this distinction matters because policy should target not only large emitters, but also categories that perform worse than expected.

Within this framework, Police Stations emerge as a plausible priority category because they combine above-expected carbon intensity with relatively high total emissions. By contrast, Recreation appears better characterized as large but relatively efficient: its total emissions are substantial, but its adjusted carbon intensity is not unusually high. Figure 3, therefore, helps distinguish between scale-driven burden and more structural decarbonization challenges.

Figure 3. Adjusted Excess GHG Intensity vs Total Emissions



4.4 Within-Type Inefficiencies and Building-Level Variation

Category-level patterns do not fully explain where inefficiencies occur. Residual analysis shows substantial **within-type heterogeneity**, meaning that buildings in the same category can still perform very differently after controlling for size, operating hours, and year. Buildings such as **Pelmo Park** and **Wellesley Community Centre** consistently underperform relative to model expectations, indicating that some inefficiencies are **facility-specific** rather than type-specific.

This suggests that category-level differences alone are not sufficient for action. Building type helps identify where the City should look first, but persistent residual underperformance indicates where intervention may matter most. A practical implication is that category-level prioritization should be combined with targeted audits, monitoring, and retrofit measures for persistently underperforming buildings.

4.5 Key Takeaways for Toronto's Building Decarbonization

This study shows that **energy efficiency** and **carbon performance** are not interchangeable across Toronto's municipal building portfolio. After controlling for floor area, operating hours, and year, raw differences in energy use intensity are substantially reduced, while more meaningful differences remain on the **carbon** dimension. The main policy-relevant variation, therefore, lies less in overall energy use than in **carbon intensity after adjustment**.

The mechanism suggested by these results is that decarbonization depends not only on the amount of energy consumed but also on the **composition of energy use**. Buildings can have similar adjusted energy performance while still differing in carbon intensity because of greater reliance on fossil-fuel-based systems, especially heating. This helps explain why **Fire Stations** remain particularly important on the carbon track. For Toronto, the implication is that efficiency improvements alone are insufficient; electrification, fuel switching, and targeted intervention for persistently underperforming facilities are also required.

5. References

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6. Supplementary Information

6.1 Convergence diagnostics, posterior predictive checking

Core models checked	E3 random-slope energy model and C3 random-slope carbon model
Diagnostics covered	Prior choice rationale, convergence diagnostics (Rhat, ESS, trace plots, and sampler warnings), posterior predictive checking, and interpretation of what was learned
Main materials reflected	Updated annotated R script, E3/C3 trace outputs, E3/C3 PPC outputs, revised caterpillar figures, and the current figure-caption direction
Benchmark kept in appendix	E2/C2 remain useful benchmark models for comparison, but the updated main-text focus is E3/C3
Bottom line	The E3/C3 models have Rhat values extremely close to 1 and trace plots that look broadly well mixed; PPC suggests they reproduce the overall centre and broad shape of the observed distributions reasonably well. However, a small number of divergent transitions remain, so the write-up should present the models as usable but not flawless.

Convergence diagnostics

Convergence diagnostics assess whether the Markov chains converged to the same posterior distribution rather than drifting, separating, or becoming stuck. Convergence was evaluated using Rhat values, effective sample sizes, and trace plots.

Rhat interpretation

Rhat compares within-chain and between-chain variation and should be close to 1 when convergence is achieved. A common rule of thumb is that values below 1.01 indicate acceptable convergence. In the updated E3 model, fixed-effect Rhat values are approximately 1.0003, 1.0025, 1.0007, and 1.0011; in the updated C3 model, they are approximately 1.0004,

0.9998, 1.0007, and 1.0012. All are very close to 1, indicating no meaningful chain-level disagreement for the key coefficients.

Posterior summaries for the main coefficients are numerically stable enough to interpret. At the same time, low Rhat does not by itself guarantee that the model is perfectly specified or free of sampling pathologies.

Trace plot interpretation

Trace diagnostics are evaluated for the E3 and C3 models. For E3, the monitored parameters include the random-slope standard deviation, the intercept–slope correlation, and the residual standard deviation (sigma). Across all parameters, the chains fluctuate within stable ranges and exhibit frequent overlap across chains, forming the characteristic “caterpillar” pattern of well-mixed posterior samples. A similar pattern is observed for C3, with chains remaining bound, heavily interwoven, and without evidence of long-run drift or persistent separation.

Compared to the simpler benchmark models (E2/C2), the E3/C3 diagnostics remain supportive overall but require a more cautious interpretation due to the presence of a small number of divergent transitions.

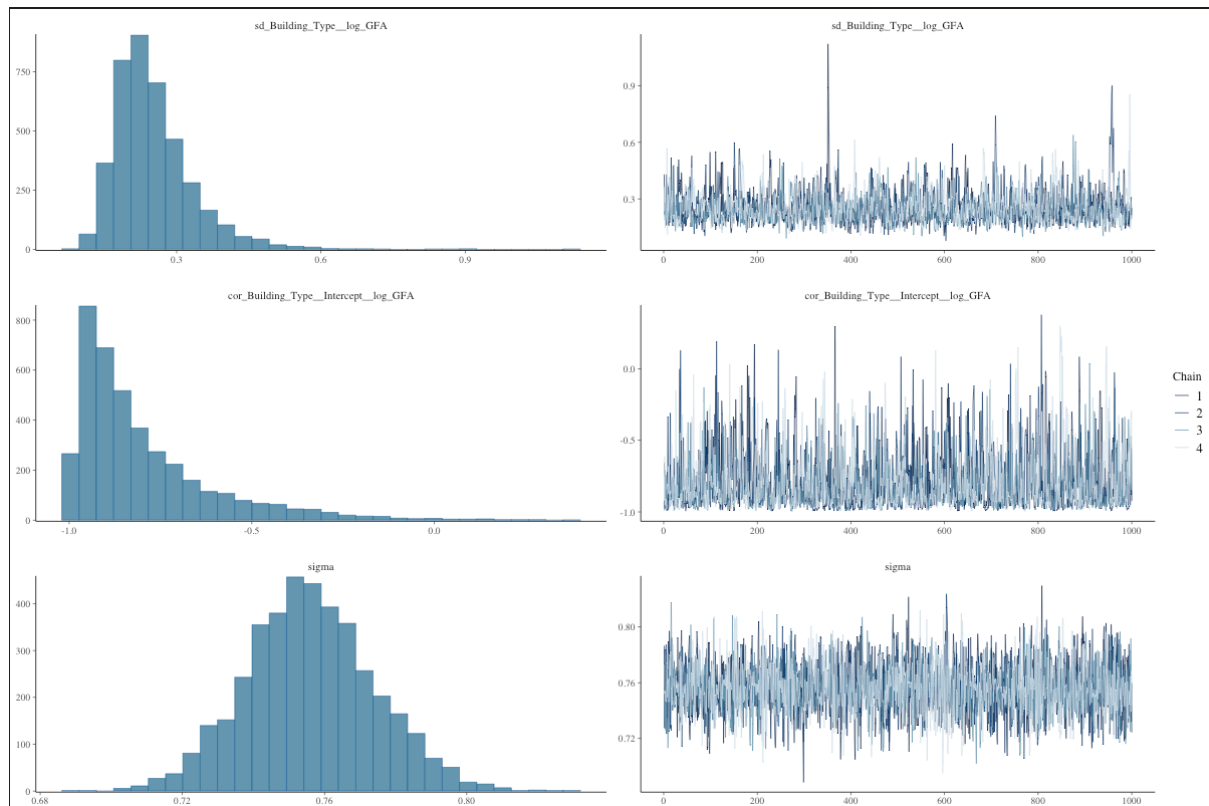


Figure A1. E3/C3 trace diagnostic.

Posterior predictive checking

The model is fit by comparing the observed data with replicated data drawn from the posterior predictive distribution. A good model should capture the key features of the observed distribution, including its center and overall shape.

For the E3/C3 models, the PPC results indicate that both the energy and carbon models capture the overall location and broad distributional shape of the observed outcomes reasonably well. In both panels, the observed density lies within the range of replicated densities, indicating that the models successfully recover the main signal in the data.

However, the fit is not perfect. The observed distributions remain more sharply peaked than the replicated curves, and some mismatch persists in the tails. This suggests that while the E3/C3 models are suitable for comparative analysis and interpretation, they smooth over certain extreme observations and finer distributional features.

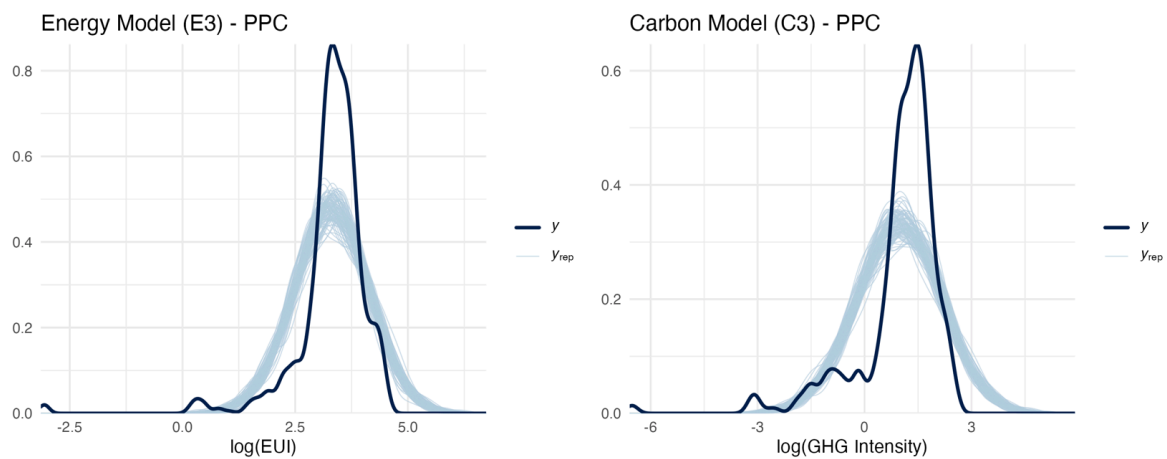


Figure A2. Posterior predictive checks for the main E3 and C3 models.

6.2 Figure Output

Figure A3. Prior Predictive Check

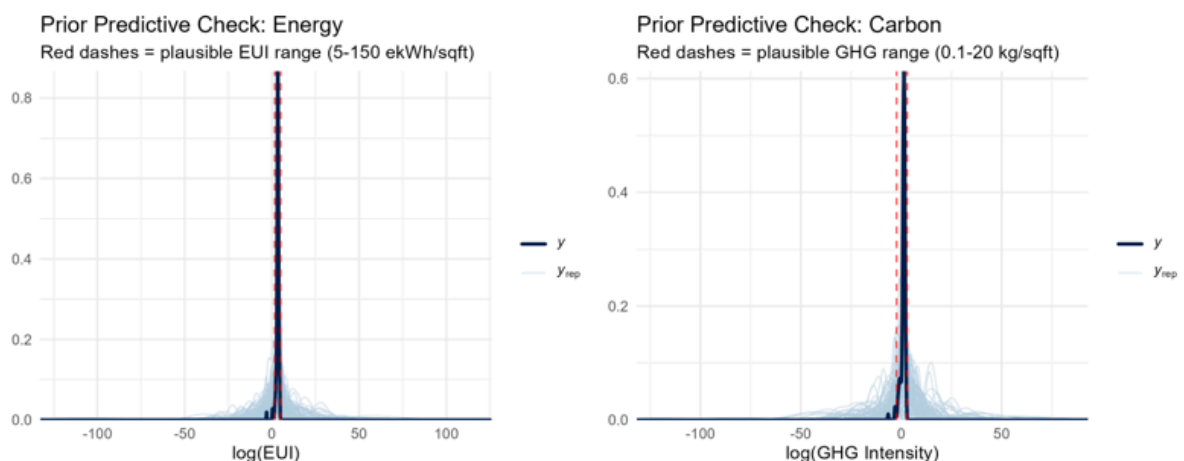


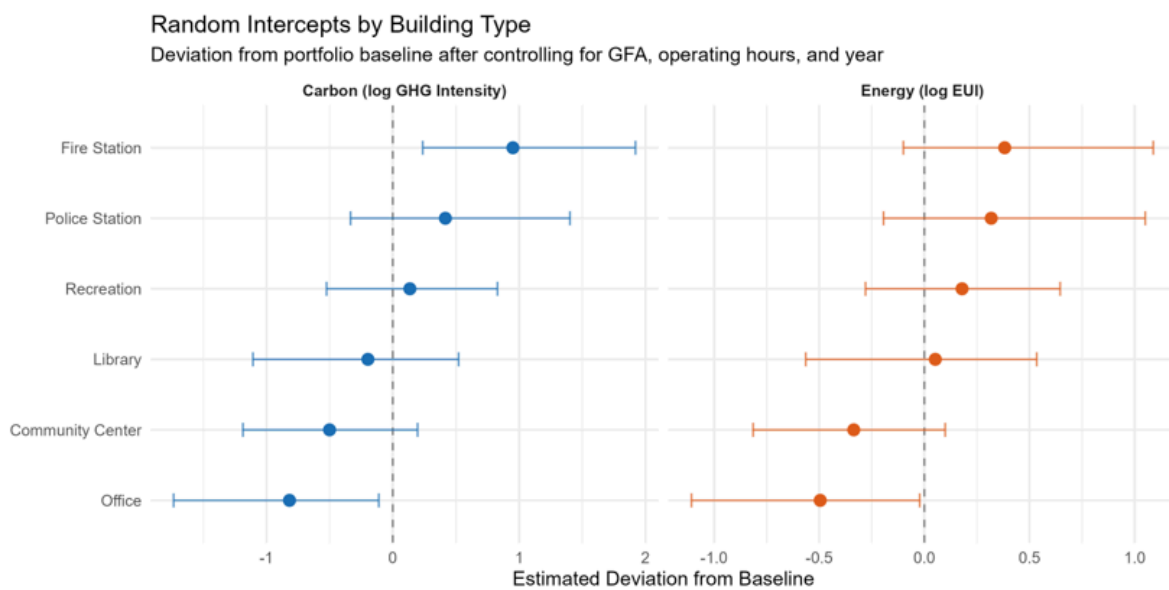
Figure A3 displays the prior predictive checks for both energy and carbon models. The generated distributions confirm that our weakly informative priors produce physically plausible ranges for EUI and GHG intensity prior to observing the data.

Table A1. LOO-CV Model Comparison

Model	elpd_diff	se_diff	elpd_loo	se_elpd_loo	p_loo	se_p_loo	looic	se_looic	Track
E3	0	0	-1055.062328	75.88433422	27.83973605	6.486156945	2110.124657	151.7686684	Energy
E2	-26.71145431	12.39908953	-1081.773783	74.42123789	19.93906843	5.268891321	2163.547565	148.8424758	Energy
E1	-62.79875349	11.21889472	-1117.861082	74.01504285	14.13534391	4.465312865	2235.722164	148.0300857	Energy
C3	0	0	-1358.714536	55.10351025	22.16845938	3.508035532	2717.429072	110.2070205	Carbon
C2	-44.99352574	14.50780808	-1403.708062	53.03805758	15.63394161	2.741128673	2807.416123	106.0761152	Carbon
C1	-80.24869497	13.99189213	-1438.963231	53.32414017	9.650265533	2.165060313	2877.926462	106.6482803	Carbon

This table documents model selection across the baseline, random-intercept, and random-slope specifications. It supports the choice of E3/C3 as the primary models by showing that the varying-intercept, varying-slope specifications provide the strongest out-of-sample predictive performance.

Figure A4. Random Intercepts by Building Type (E2/C2)



This appendix figure reports the simpler random-intercept benchmark models (E2/C2), in which building types are allowed to differ in baseline level but are assumed to share a common floor-area slope. Relative to the main E3/C3 specifications, these models provide a more constrained comparison and are included as a supplementary reference rather than the primary basis for interpretation.

The E2/C2 results remain useful for showing the broad direction of type-level differences under a simpler specification. In particular, Fire Station already appears above baseline—especially on the carbon track—while several other categories remain closer to zero or less clearly separated. However, compared with E3/C3, the E2/C2 models do not

capture the possibility that the effect of floor area varies across building types, so they should be interpreted as a benchmark rather than the main substantive result.

6.3 Overall Descriptive Statistics

Variable	N	Mean	SD	Median	Min	Max
Gross Floor Area (sqft)	914	40,509.02	79,742.84	12,718.00	1,032.00	787,186.00
Weekly Operating Hours	914	111.57	37.59	100.00	52.00	168.00
EUI (ekWh/sqft)	914	33.03	19.53	30.07	1.27	97.02
GHG Intensity (kg CO2e/sqft)	914	3.83	2.53	3.52	0.08	13.47

Sample: Toronto municipal building panel, 2017–2024 (excluding 2021)

6.4 Descriptive Statistics by Building Type

Building Type	N	Mean GFA (sqft)	SD GFA	Mean Hours	SD Hours	Mean EUI	SD EUI	Mean GHG Intensity	SD GHG Intensity
Community Centre	124	23,335.11	28,075.32	108.55	35.70	32.95	20.19	3.27	2.78
Fire Station	235	12,358.97	4,337.29	147.97	26.29	44.25	18.44	4.83	2.06
Library	216	28,896.12	27,031.51	93.52	13.56	29.59	15.75	3.49	2.03
Office	150	87,534.81	153,304.15	76.24	14.72	22.29	16.36	2.33	1.80
Police Station	74	29,676.04	23,822.95	128.11	45.45	28.97	16.90	3.06	2.01
Recreation	115	47,942.84	66,493.33	112.80	28.67	46.36	22.20	5.47	3.55

AI Use Statement

AI tools were used only for editing support and limited debugging assistance in this project, including minor help with parts of the coding/data-cleaning workflow, language polishing, and concision editing. All data processing, model specification, Bayesian estimation, diagnostic checks, statistical outputs, and empirical interpretations were completed, verified, and can be fully explained by the authors. No AI tool was used to generate or automate the reported statistical results, figures, tables, or substantive findings. The authors take full responsibility for all content in this paper.